

tRNA research 四个子项目资料汇报

基于 PubMed 与 GitHub 的资料筛选、文献高亮、图像证据与研究路线整合

白底黑字版 | 2026-05-23 | 中文主讲

四个文件夹：结构功能、工具算法、修饰指纹数据库、湿实验

资料筛选策略

PubMed literature plus GitHub resources

PubMed：每个主题保留 15 篇 2020-2026 年候选文献，优先高影响力期刊；不足时按题名 / 摘要相关性补齐。

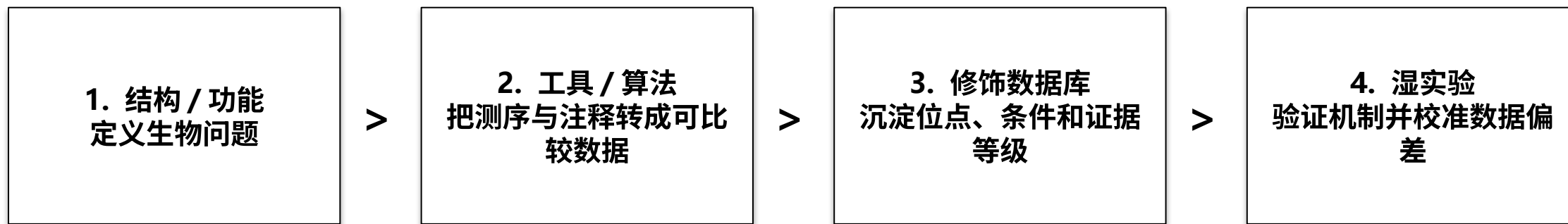
GitHub：按文件夹主题关键词检索仓库，保存 stars、语言、更新时间、许可证与链接；去除了 transformation/transformer 等拼写噪音。

图像：优先抓取开放 PMC 文章页中的原始 figure 图像；每张图均在 PPT 里标注 PMID/PMCID/ 期刊来源。

限制：未调用 Clarivate JCR 官方数据库，因此“影响因子最高”在本资料包中体现为高影响力期刊代理排序。

四个文件夹形成一条 tRNA 研究链路

From folders to an integrated tRNA research workflow



建议把四个子项目视为一个闭环：文献提出假设，工具产生特征，数据库固化证据，湿实验验证因果。

PPT 中所有原始图均来自 PMC 开放页面或保存的文章图像文件，并在图下标出来源。

tRNA 的生物功能与结构特征

Biological functions and structural characteristics of tRNA

中文阐述

tRNA 不只是“接头分子”，而是连接遗传密码、翻译速度、细胞应激、免疫防御和疾病治疗的调控节点。

结构层面从三叶草二级结构、L 型三级结构、修饰核苷到氨酰化 / 核糖体复合物共同决定读取精度。

近期高影响力研究集中在 suppressor tRNA、tRNA 尾部切割、tRNA 片段和 tRNA 结构互作组。

English explanation

tRNA functions as more than an adaptor: it links genetic decoding, translation kinetics, stress response, immunity, and therapeutic engineering.

Its cloverleaf secondary structure, L-shaped tertiary fold, modified nucleosides, and aminoacylation state jointly determine decoding fidelity.

Recent high-impact work emphasizes suppressor tRNAs, tRNA-tail cleavage, tRNA-derived fragments, and in vivo tRNA structurome/interactome maps.

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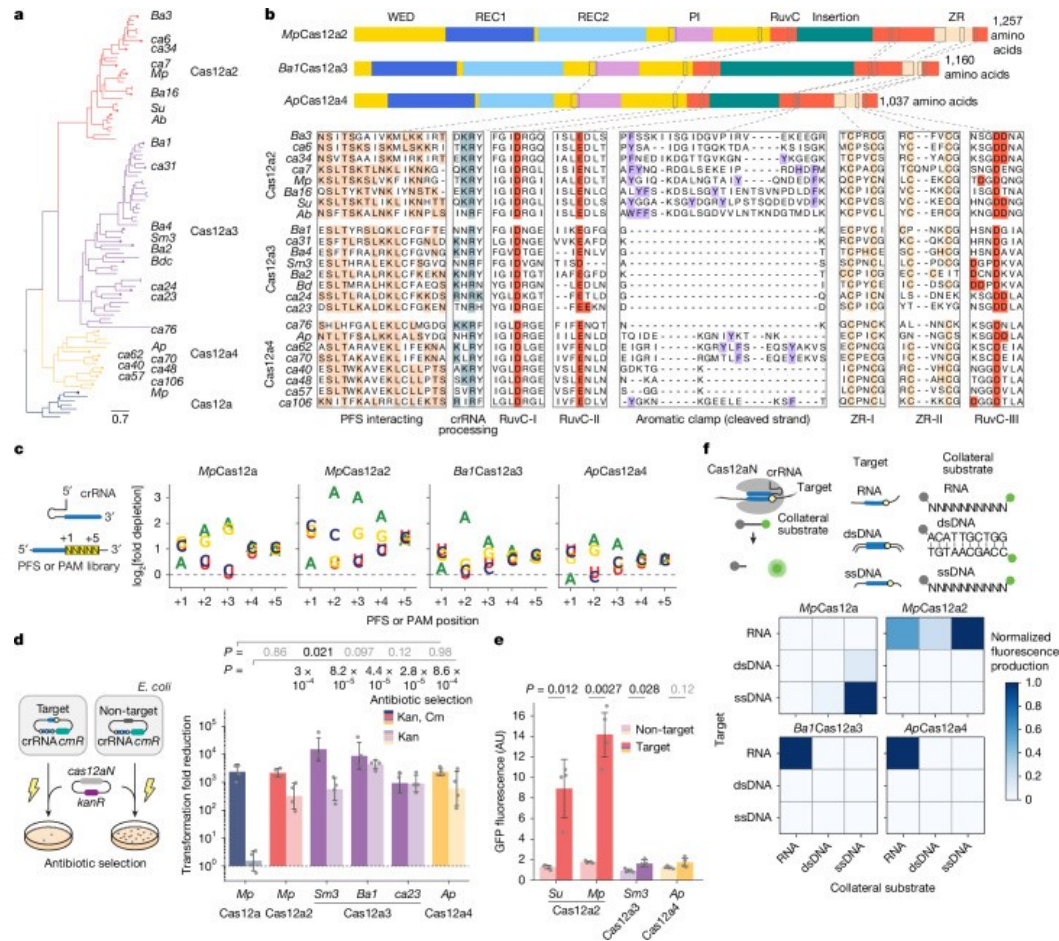
生物功能与结构：最新高影响力文献清单

Top recent high-impact PubMed records

#	题名	期刊 / 年份	PMID/DOI
1	RNA-triggered Cas12a3 cleaves tRNA tails to execute bacterial immunity.	Nature / 2026	41501459 10.1038/s41586-025-09852-9
2	A virally encoded tRNA neutralizes the PARIS antiviral defence system.	Nature / 2024	39111359 10.1038/s41586-024-07874-3
3	tRNA modifications tune m6A-dependent mRNA decay.	Cell / 2025	40311619 10.1016/j.cell.2025.04.013
4	An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery.	Nature biotechnology / 2026	41555020 10.1038/s41587-025-02982-5
5	AAV-delivered UGA suppressor tRNA for disease-agnostic in vivo gene therapy.	Nature biotechnology / 2026	41629463 10.1038/s41587-026-02999-4
6	tRNA as an assembly chaperone for a macromolecular transcription-processing complex.	Nature structural & molecula / 2025	40908366 10.1038/s41594-025-01653-y
7	Structural basis of tRNA recognition by the m3C RNA methyltransferase METTL6 in complex wi	Nature structural & molecula / 2024	38918637 10.1038/s41594-024-01341-3
8	A methyltransferase-independent role for METTL1 in tRNA aminoacylation and oncogenic trans	Molecular cell / 2025	39892392 10.1016/j.molcel.2025.01.003
9	Translational regulation by oxidative desulfuration of tRNA modifications.	Nature communications / 2026	41807381 10.1038/s41467-026-70126-7
10	Structure of fungal tRNA ligase Trl1 with RNA reveals conserved substrate-binding principl	Nature structural & molecula / 2025	40563009 10.1038/s41594-025-01589-3

生物功能与结构：文章图像证据

Representative figure from screened literature



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仓库	stars	语言	用途判断
junchaoshi/sports1.1	57	Perl	Small non-coding RNA annotation Pipeline Optimized for rRNA- and tRNA-Derived Smal
UCSC-LoweLab/tDRnamer	2	Python	Standardized naming for tRNA-derived RNAs

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tRNA 分析工具与算法

tRNA analysis tools and algorithms

中文阐述

算法核心难点来自 tRNA 的短序列、多拷贝、强修饰、结构相似性以及测序逆转录停顿 / 错配。

工具链通常包括基因识别、参考库构建、读段比对、修饰信号建模、表达定量和可视化解释。

tRNAscan-SE、mim-tRNAseq、nanopore direct tRNA sequencing 等代表了从注释到定量修饰的关键路线。

English explanation

Algorithmic challenges arise from short length, multi-copy genes, heavy modification, structural similarity, and modification-induced RT stops or mismatches.

A practical toolchain covers gene detection, reference construction, read alignment, modification-signal modeling, abundance quantification, and visualization.

tRNAscan-SE, mim-tRNAseq, and direct nanopore tRNA sequencing represent major routes from annotation to quantitative modification profiling.

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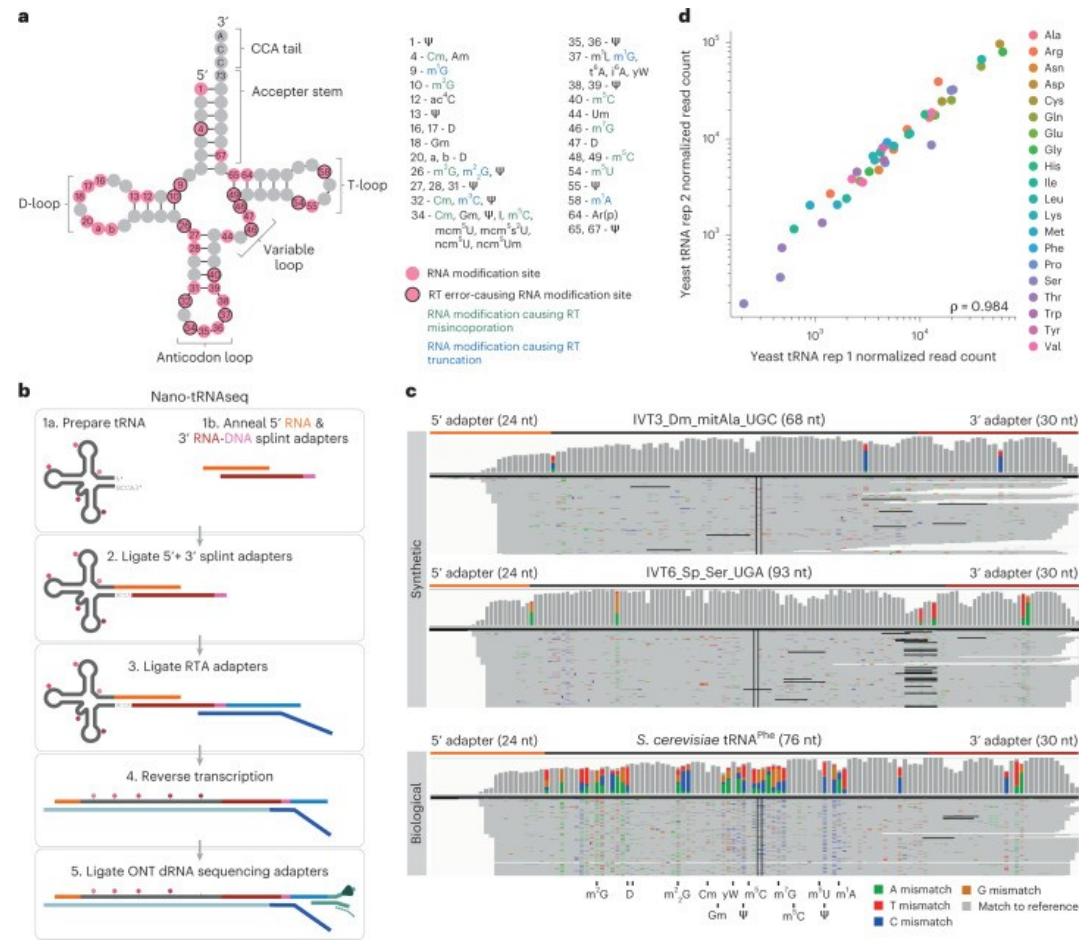
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3	Design, construction, and functional characterization of a tRNA neochromosome in yeast.	Cell / 2023	37944512 10.1016/j.cell.2023.10.015
4	Quantitative analysis of tRNA abundance and modifications by nanopore RNA sequencing.	Nature biotechnology / 2024	37024678 10.1038/s41587-023-01743-6
5	Quantifying tRNA abundance by sequencing.	Nature genetics / 2023	37173525 10.1038/s41588-023-01404-z
6	High-resolution quantitative profiling of tRNA abundance and modification status in eukary	Molecular cell / 2021	33581077 10.1016/j.molcel.2021.01.028
7	A ligation-independent sequencing method reveals tRNA-derived RNAs with blocked 3' termini	Molecular cell / 2024	39096899 10.1016/j.molcel.2024.07.008
8	A pro-metastatic tRNA fragment drives Nucleolin oligomerization and stabilization of its b	Molecular cell / 2022	35654044 10.1016/j.molcel.2022.05.008
9	Deciphering tRNA repertoires and translation coordination during mouse early embryogenesis	Nature communications / 2026	42062277 10.1038/s41467-026-72603-5
10	In vivo structure profiling reveals human cytosolic and mitochondrial tRNA structurome and	Nature communications / 2025	40447571 10.1038/s41467-025-59435-5

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Source: Quantitative analysis of tRNA abundance and modifications by nanopore RNA sequencing.; Nature biotechnology, 2024; PMID 37024678; PMCID PMC10791586

工具与算法： GitHub 与可复用资源

GitHub repositories and reusable assets

仓库	stars	语言	用途判断
UCSC-LoweLab/tRNAscan-SE	89	C	A program for detection of tRNA genes
nediaalkova-lab/mim-tRNAseq	24	Scheme	Modification-induced misincorporation tRNA sequencing
merenlab/tRNA-seq-tools	11	Python	无描述
rnabioco/tRNA004	8	Python	Nanopore direct tRNA sequencing method refinement and RNA modification benchmarkin
rnabioinfor/TRAC-Seq	5	HTML	Single-nucleotide resolution profiling of m7G tRNA modifications
UCSC-LoweLab/tDRnamer	2	Python	Standardized naming for tRNA-derived RNAs

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tRNA 修饰指纹数据库

tRNA mod-fingerprint database

中文阐述

tRNA 修饰指纹把“哪一种 tRNA、哪个位点、哪类修饰、在哪个条件下变化”组织为可查询的证据结构。

数据库建设需要统一 tRNA 坐标、修饰命名、物种来源、实验技术、置信度和可视化字段。

MODOMICS、RNAcentral 及测序 / 质谱分析项目可作为设计本地数据库的外部锚点。

English explanation

A tRNA modification fingerprint organizes which tRNA, which position, which modification, and which condition into a queryable evidence model.

Database design requires harmonized tRNA coordinates, modification nomenclature, species metadata, experimental technology, confidence scores, and visualization fields.

MODOMICS, RNAcentral, and sequencing/mass-spectrometry projects can anchor a local database schema.

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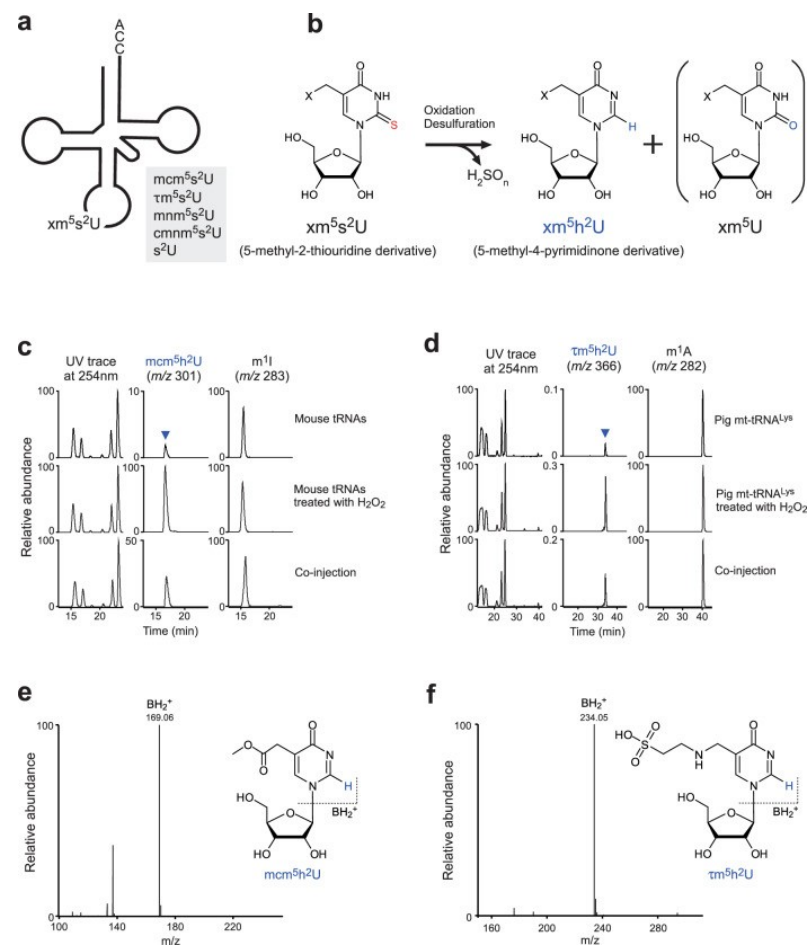
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2	Alanyl-tRNA synthetase, AARS1, is a lactate sensor and lactyltransferase that lactylates p	Cell / 2024	38653238 10.1016/j.cell.2024.04.002
3	A cholesterol-responsive hepatic tRNA-derived small RNA regulates cholesterol homeostasis	Nature communications / 2025	41398161 10.1038/s41467-025-67387-z
4	Transfer RNA modifications and cellular thermotolerance.	Molecular cell / 2024	38181765 10.1016/j.molcel.2023.11.041
5	NAT10 promotes cancer metastasis by modulating p300/CBP activity through chromatin-associa	Molecular cell / 2025	41344331 10.1016/j.molcel.2025.11.010
6	tRNA as an assembly chaperone for a macromolecular transcription-processing complex.	Nature structural & molecula / 2025	40908366 10.1038/s41594-025-01653-y
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9	Structures of the ribosome bound to EF-Tu-isoleucine tRNA elucidate the mechanism of AUG a	Nature structural & molecula / 2024	38538914 10.1038/s41594-024-01236-3
10	tRNA renovatio: Rebirth through fragmentation.	Molecular cell / 2023	37802077 10.1016/j.molcel.2023.09.016

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jfallmann/tRNA_ngs_mod_map_call_galaxy	3	NA	Tool collection for identification of tRNA modifications from NGS data galaxy work
monimaanam/Modomics_Decoder	1	Python	Annotates modomics RNA sequences for easy viewing of RNA modifications and nucleot
AteeshaNegi/nanopore-2Read-trna-pipeline	0	NA	Oxford Nanopore 2-Read Genome Wide tRNA Modification Analysis Pipeline - RNA Epitr
FlorianPichot/tRNA_reference_construction	0	R	无描述

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tRNA 相关湿实验

tRNA-related wet-lab experiments

中文阐述

湿实验围绕样本质量、tRNA 富集、修饰保真、氨酰化状态保护和测序 / 质谱平台兼容性展开。

常见实验包括 Northern blot、氨酰化测定、LC-MS/MS、去修饰 / 酶处理、small RNA-seq 与 nanopore direct RNA sequencing。

关键质控包括 spike-in、去氨酰化对照、酶切效率、RT-stop/mismatch 校准以及 tRNA 片段与成熟 tRNA 的区分。

English explanation

Wet-lab work depends on sample integrity, tRNA enrichment, modification preservation, aminoacylation protection, and platform compatibility.

Common assays include Northern blotting, aminoacylation assays, LC-MS/MS, demodification/enzyme treatment, small RNA-seq, and direct nanopore RNA sequencing.

Key controls include spike-ins, deacylation controls, enzyme-efficiency checks, RT-stop/mismatch calibration, and discrimination between tRNA fragments and mature tRNAs.

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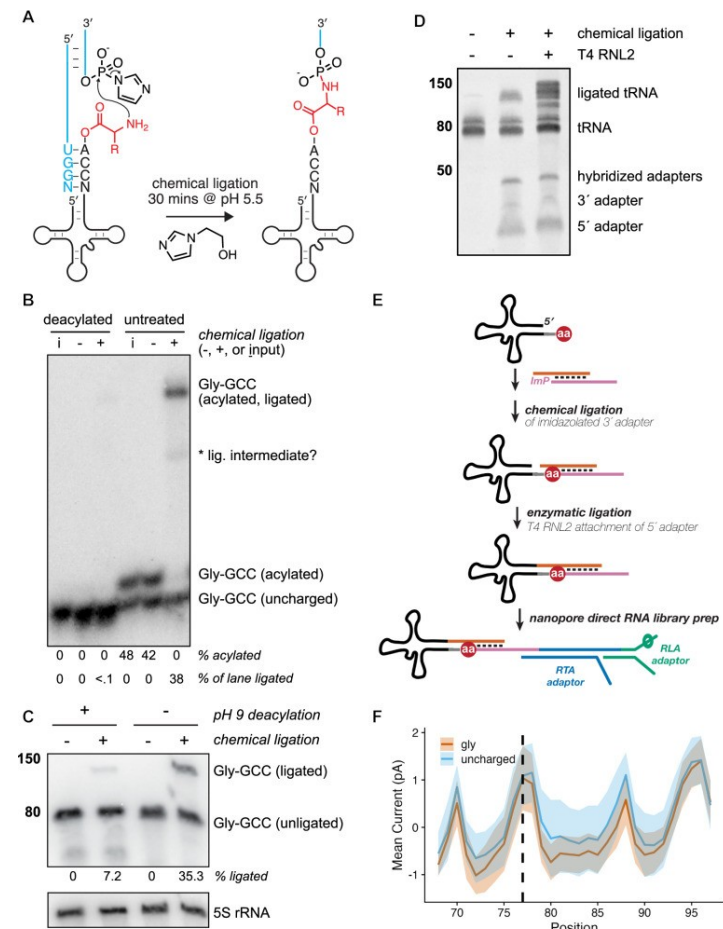
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8	RelA-SpoT Homolog toxins pyrophosphorylate the CCA end of tRNA to inhibit protein synthesis	Molecular cell / 2021	34174184 10.1016/j.molcel.2021.06.005
9	A ligation-independent sequencing method reveals tRNA-derived RNAs with blocked 3' termini	Molecular cell / 2024	39096899 10.1016/j.molcel.2024.07.008
10	tRNA m1A modification regulate HSC maintenance and self-renewal via mTORC1 signaling.	Nature communications / 2024	38977676 10.1038/s41467-024-50110-9

湿实验路线：文章图像证据

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rnabioinfor/TRAC-Seq	5	HTML	Single-nucleotide resolution profiling of m7G tRNA modifications
rnabioco/aa-tRNA-seq	2	Python	Public-facing repo for aa-tRNA-seq pre-print
tryptchromics/trnaExpCodAdapt	0	R	tRNA Expression-Codon Adaptation

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综合结论：四个子项目应合并为证据闭环

Integrated conclusion

结构功能主题提供生物学假设：哪些 tRNA、修饰或片段可能改变翻译、免疫或疾病表型。

工具算法主题把高通量数据转成可比较特征：丰度、错配、RT-stop、修饰概率、结构互作与片段谱。

修饰指纹数据库主题负责证据沉淀：统一位点坐标、文献来源、实验方法、置信度和查询接口。

湿实验主题负责验证：通过 LC-MS/MS、Northern、氨酰化测定和定向测序校准算法输出。

后续建议

Suggested next steps

如果要继续深化，建议先确定一个核心物种 / 疾病 / 实验条件，否则四个主题会很容易扩散。

数据库字段优先级：tRNA ID、isoacceptor/isodecoder、修饰位点、修饰类型、检测技术、PMID、样本条件、置信度。

实验优先级：先做可重复的 tRNA abundance/modification profiling，再针对关键位点做验证实验。

文献更新：建议每月按相同脚本刷新 PubMed 与 GitHub，避免综述和工具版本过期。